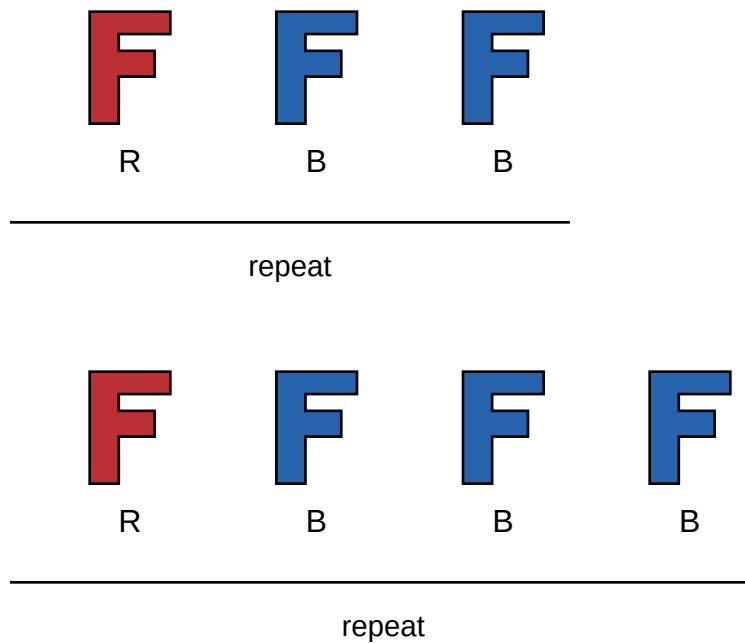
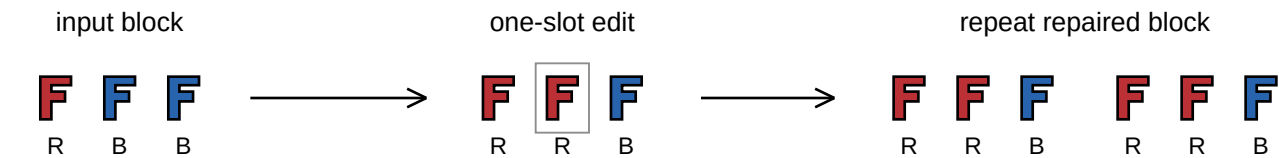


Borders continue endlessly in both directions. Only the footprints and their colors count. A slide must match every footprint in both rows; rows keep their places.

Problem 1: Repeat each row below on a long strip. Find the shortest slide that matches both rows together. Design a different pair with shortest shared slide 6 slots, and another with shortest shared slide 8 slots. Neither row may match after a one-slot slide.



Problem 2: Recolor the fewest footprints in each displayed block, then repeat the same repair endlessly so its requested slide becomes a match. Keep positions and forms fixed. Find equally short repairs when possible, and explain why fewer changes cannot work.



slide 3 slots



slide 2 slots

changes: _____



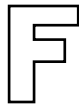
slide 2 slots

changes: _____

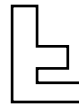


changes: _____

Use the two mirror forms below. Card edges are only holders; forms count.



A



mirror A

Problem 3: On the table, make necklaces with 5, 6, 7 and 8 cards. Every pair of neighbors, including the last and first, must use opposite forms. Which can be finished? For the others, make as few equal-neighbor joins as possible. Record the forms in the circles.

